Oscar A. Flores-Gaitán

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Education

Ph.D. in Astronomy & Astrophysics — *Newark, DE*
Department of Physics & Astronomy, University of Delaware — Aug 2026 – Present
- Advisor: Prof. Sarah Dodson-Robinson

Licentiate in Physics — *Guatemala City*
Department of Physics, Universidad del Valle de Guatemala — Jan 2022 – May 2026
- Honors: Magna Cum Laude
- Thesis: "*Decoding the Barnard's Star Planetary System: A Time Series Based Analysis of the Planetary Architecture*" (Advisor: Prof. Sarah Dodson-Robinson)

Research Interests

- Precision Doppler detection and characterization of exoplanetary systems.
- Modeling and mitigating stellar activity in radial velocity and spectroscopic time series.
- Developing and applying machine learning and statistical techniques to astronomical data.

Publications

First-Author

2. **Flores-Gaitán, O.A.**, Dodson-Robinson, S., *"Detecting Harmonic Coupling in Stellar Activity using Dual-Frequency Autocoherence"*. [Manuscript in preparation, 2027].
1. **Flores-Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., *"A Dynamically Stable Three-Planet Architecture for Barnard's Star"*. [To be submitted, 2026].

Second- & Third-Author

1. Cambronero Jiménez, I., Dodson-Robinson, S., **Flores-Gaitán, O.A.**, *"Cepstrum Analysis of Stellar Activity"*. [Manuscript in preparation, 2027].

Co-Authored & Contributor

2. Zhao, L.L. et al., **including Flores-Gaitán, O.A.**, *"The Extreme Stellar-Signals Project IV"*. [Manuscript in preparation, 2026].
1. Dressing, C.D., Adams, D., Alecian, E. et al., **including Flores-Gaitán, O.A.**, *"Early Exploration of the Scientific Discovery Space for the Habitable Worlds Observatory"*. arXiv preprint (2026). arXiv:2608.11294.

Theses

1. **Flores-Gaitán, O.A.**, *"Decoding the Barnard's Star Planetary System: A Time Series Based Analysis of the Planetary Architecture (In Spanish)"*. Licentiate Thesis, Universidad del Valle de Guatemala (2026). Repository Handle.

Software

1. **Flores-Gaitán, O.A.**, *"orbcloud: 3D orbital probability clouds from simulated exoplanet posteriors"*. Zenodo (2026). DOI: 10.5281/zenodo.20935725 GitHub.
- Python library to generate 3D orbital probability clouds from exoplanet MCMC posteriors.

Talks & Posters

Contributed & Invited Talks

4. **Flores-Gaitán, O.A.**, *"A Dynamically Stable Three-Planet Architecture for Barnard's Star"*, **Talk** at Exoplanet Group Meeting, Department of Physics & Astronomy, University of Delaware, Newark, DE, October 2026.

3. **Flores-Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., Rubio-Herrera, E., *"How many terrestrial planets orbit Barnard's Star?"*, **Contributed Talk** at APS Mid-Atlantic Section Meeting 2025, Penn State Harrisburg, Middletown, PA, November 15, 2025. Link
2. **Flores-Gaitán, O.A.**, *"Hunting for Exoplanets: Real Planets or Stellar Processes?"*, **Invited Talk** at Universidad del Valle de Guatemala's Physics Week, Guatemala City, October 25, 2025.
1. **Flores-Gaitán, O.A.**, *"Detecting and Characterizing Exoplanets with TESS"*, **Invited Talk** at Universidad del Valle de Guatemala's Astronomy Club, Guatemala City, May 9, 2025.

Conference Posters

3. **Flores-Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., *"Disentangling Planetary and Stellar Signals in Barnard's Star"*, **Poster** at Physics Science Fair at Universidad del Valle de Guatemala's Physics Week, Guatemala City, October 24, 2025.
2. **Flores-Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., *"Disentangling Planetary and Stellar Signals in Barnard's Star"*, **Virtual Poster** at NASA's Towards the Habitable Worlds Observatory: Visionary Science and Transformational Technology, Johns Hopkins University, Washington, DC, July 28-31, 2025.
1. **Flores-Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., *"Disentangling Planetary and Stellar Signals in Barnard's Star"*, **Virtual Poster** at Sagan Summer Workshop, NASA Exoplanet Science Institute, California Institute of Technology, Pasadena, CA, July 23, 2025. Link

Research Experience

Graduate Researcher *Newark, DE*
Department of Physics & Astronomy, University of Delaware Aug 2026 – Present
Advisor: Prof. Sarah Dodson-Robinson

- Analyzing solar observations to test and calibrate spectral coherence methods for measuring stellar rotation and magnetic activity cycles.
- Developing dual-frequency coherence methods to disentangle stellar activity from planetary signals in radial velocity (RV) datasets.

Undergraduate Visiting Researcher *Newark, DE*
Department of Physics & Astronomy, University of Delaware Jun 2025 – Jul 2026
Advisor: Prof. Sarah Dodson-Robinson

- Evaluated advanced statistical methods to distinguish genuine exoplanetary signals from stellar noise in high-precision RV data.
- Implemented gaussian process (GP) regression and frequency-domain analysis to mitigate stellar variability and improve the reliability of RV detections.
- Analyzed 25 years of multi-instrument RV datasets to determine the planetary architecture of Barnard's Star, utilizing advanced statistical metrics for model validation.

Undergraduate Researcher *Guatemala City*
Department of Physics, Universidad del Valle de Guatemala Nov 2024 – May 2025
Advisor: Prof. Eduardo Rubio-Herrera

- Processed and analyzed photometric time series data from the TESS mission to identify exoplanet candidates.
- Developed an algorithm to detect transits and extract orbital parameters from stellar light curves.

Academic Appointments & Teaching

Graduate Teaching Assistant *Newark, DE*
Department of Physics & Astronomy, University of Delaware Aug 2026 – Present

- 4 course sections: PHYS 221 (Introductory Physics Laboratory I).

Undergraduate Teaching Assistant & Laboratory Assistant *Guatemala City*
Physics Department, Universidad del Valle de Guatemala Jan 2024 – May 2025

- 7 course sections: Physics I (FF2016), Physics II (FF2017), Physics III (FF2018), and Classical Mechanics I (FF3027).

Academic Tutor *Guatemala City*
Universidad del Valle de Guatemala Jun 2022 – Present

- Provided 350+ hours of individual tutoring for high school and university students in Calculus I–III, Differential Equations, and Physics I–III.

Industry & Professional Experience

Data & Research Analyst *Guatemala City*
Vekst Consulting Group Feb 2026 – May 2026

- Led scientific consulting and quantitative modeling projects, applying advanced statistical methods to financial and market data analysis.
- Transformed rigorous research into high-value strategic assets and technical documentation for the consulting sector.
- Designed, developed, and deployed the firm's official website, expanding technical and commercial visibility.

Mentorship

Isabel Cambronero Jiménez (Universidad de Costa Rica) *Newark, DE*
University of Delaware Undergraduate Research Program Jun 2026 – Present
Primary Advisor: Prof. Sarah Dodson-Robinson

- Project: Application of the Cepstrum Technique to Exoplanet Radial Velocity Time Series.

Workshops & Summer Schools

Cohere Labs Machine Learning Summer School *Remote*
Cohere Labs Aug 2026

Code/Astro Workshop *Santa Cruz, CA*
University of California, Santa Cruz Jun 2026

Alpha-Cen Astronomy Summer School *Remote*
Alpha-Cen (Central American and Caribbean Astrophysics) Aug 2025

Workshop on Astrophysics of High Energies and Cosmology *Guatemala City*
Universidad Nacional Autónoma de México (UNAM) Nov 2024

Fellowships, Grants & Awards

- **Code/Astro Full Financial Aid Grant**, University of California, Santa Cruz: Awarded full financial support of $1,180 for scientific software development workshop (2026).
- **Magna Cum Laude**, Universidad del Valle de Guatemala: Graduated with high academic honors in Physics (2026).
- **Student Travel Award**, University of Delaware: Awarded $1,000 travel grant to present research at an international conference (2025).
- **Summer Visiting Research Fellowship**, Department of Physics & Astronomy, University of Delaware: Selected out of 147 international applicants (2025).
- **3x Distinguished Student Award** (*Estudiante Distinguido*), Universidad del Valle de Guatemala: Awarded for top academic standing (2022, 2023, 2024).
- **Experimental Physics Record**, Universidad del Valle de Guatemala: Achieved the university's historically most accurate experimental measurement of local gravitational acceleration ($g = 9.7876 \pm 0.0026$ m/s^2) (2024).
- **Top 10 Finalist**, Guatemalan National Mathematics Olympiad (2020).

Technical Skills

Programming Languages: Python, LaTeX, R (Intermediate)

Data Analysis & Probabilistic Modeling: Statistical Inference, Gaussian Processes (GPs), Markov Chain Monte Carlo (MCMC), Bayesian Modeling, Time-Series Analysis

Machine Learning: Scikit-learn, XGBoost, Random Forests, Neural Networks, Autoencoders, K-Nearest Neighbors (KNN), Hyperparameter Tuning, Cross-Validation

Astronomy Tools: PyMC, celerite2, exoplanet, AstroPy, Lightkurve, Astroquery, ExoCTK, NWelch

Languages: Spanish (Native), English (C1), German (B2/C1)

Outreach & Leadership

- **Invited Panelist**: *"Experiences of Internships Abroad for Aspiring Guatemalan Astronomers"*, CAFE 2026 (Workshop on Astronomical Sciences: Education and Training), Universidad Rafael Landívar, Guatemala City (Jul 2026).
- **Science Fair Judge**: Evaluated undergraduate physics and engineering research projects at Universidad del Valle de Guatemala's Science Fair, Guatemala City (Jun 2026).
- **Workshop Developer & Instructor**: Designed and taught *"How to Apply to Internships in the US"* for undergraduate physics students in Latin America (Feb 2026).
- **Guest Lecturer & Teaching Assistant**: Collaborated on Extragalactic Astronomy lecture for high school students, Astro Antigua, Antigua Guatemala (Dec 2025).
- **Lead Organizer**: Physics Week 2025, a week-long conference and workshop series featuring national and international speakers, Universidad del Valle de Guatemala, Guatemala City (Oct 2025).
- **Workshop Developer & Instructor**: Designed and taught *"How to Write an Academic CV"* for first-year physics students, Universidad del Valle de Guatemala, Guatemala City (Oct 2025).
- **Active Member**: Physics Students Association (AEF-UVG), organizing public lectures and science outreach events (2025).
- **Author & Content Creator**: Authored undergraduate Physics laboratory guide and created educational laboratory videos at Universidad del Valle de Guatemala (2024). Link
- **Founder & Manager**: *Countries Kingdom*, educational geography and linguistics outreach project with over **15,000 followers** and **5,000,000+ total views** (2021 – 2023). Link

Collaborations & Working Groups

- **Extreme Stellar Signals Project (ESSP)**: Collaboration Member through the Sidera Research Group, University of Delaware.
- **Extreme Precision Radial Velocity Research Coordination Network (EPRV RCN)**: Member.

Media

3. **Actualidad UVG**, *"Oscar Flores-Gaitán demonstrates the reach of Physics in the search for exoplanets (In Spanish)"*, **Feature Article** at Universidad del Valle de Guatemala, Guatemala City, 2026. Link
2. **Ciencia UVG Podcast**, *"A UVG Physicist in Delaware (In Spanish)"*, **Podcast Episode** at Universidad del Valle de Guatemala, YouTube (Video) / Spotify (Audio), 2026.
1. **Educational Video Tutorial**, *"Measuring the gravitational acceleration g with a physical pendulum (In Spanish)"*, **YouTube Video** at Universidad del Valle de Guatemala, Guatemala City, 2024. Link